What Minimal Einstein–Cartan Torsion Does for the Bounce and Cannot Do for Dark Energy

Houston Golden[1,*]

[1] *Independent Researcher, Los Angeles, California, USA*

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Popławski's Einstein–Cartan black-hole cosmology replaces the classical singularity with a torsion-supported bounce: eliminating the non-propagating spin connection from the minimal Einstein–Cartan–Holst (ECH) action generates an effective spin–spin four-fermion contact term that is repulsive at high spin density, the same mechanism that produces Popławski's torsion-induced avoidance of gravitational singularities [1, 2]. We ask whether the identical algebraic mechanism can also source the observed late-time dark-energy density and answer, systematically, no. Consolidating two companion papers into a single Note, we (i) derive the minimal axial–axial contact interaction $-(3\kappa/16)[\gamma^2/(1+\gamma^2)]J_5^2$ from the same Cartan equation that produces the bounce term, and show its declared direct-channel, hard-cutoff, standard mean-field NJL scalar projection is repulsive, $G_s = -3\kappa/16$, so the real homogeneous gap equation for this condensate channel has no nonzero solution; (ii) prove, for canonical scalar matter, a perturbation-transparency theorem: torsion vanishes at every classical perturbation order and the Holst sector decouples identically from the scalar and tensor equations of motion, because the Holst dual contraction vanishes pointwise by the algebraic Bianchi identity on the torsion-free branch; and (iii) catalog fourteen distinct mechanism-class constraints – amplitude suppressions, naturalness deficits, decoupling identities, and classification arguments spanning seven structural foundations and six observational branches – that jointly close the four enumerated channels (NJL contact, one-loop Holst correction, Immirzi running, parity-odd CMB coupling) by which minimal ECH could plausibly connect its bounce-scale torsion to a late-time Λ-like density. A six-member spanning list of dimension-four parity-odd local densities admitted by the minimal field content is shown to have rank four as a set of operators and rank two modulo total derivatives – a generating set, not an independent basis – with every member either an exact total derivative or a Fierz-closed, $M_{\rm Pl}^{-2}$-suppressed contact term; the on-shell ECH torsion itself carries both an axial and a trace-vector irrep at finite Barbero–Immirzi parameter γ (ratio $\beta/\alpha = 1/2\gamma$; the tensor irrep vanishes identically), a correction to the purely-axial reading used in an earlier catalog draft. The result is a clean structural dichotomy: the same contact term that plausibly supports Popławski's bounce mechanism is, under minimal coupling, parity-even, Planck-suppressed, and classically transparent to perturbations – exactly the properties that make it unable to generate the late-time acceleration independently attributed to dark energy. No ECH dark-energy or birefringence prediction is made; this is a channel-level, not operator-level, closure within the stated minimal-coupling scope.

I. INTRODUCTION

Minimal Einstein–Cartan–Holst (ECH) gravity promotes the spin connection to an independent field sourced algebraically by fermion spin currents. In loop-quantized cosmology the resulting torsion term replaces the classical singularity with a quantum bounce, and in Popławski's black-hole-cosmology programme the same torsion term is the mechanism by which a collapsing fermionic interior avoids a singularity and instead bounces, producing a new expanding universe inside what an outside observer sees as a black hole [1, 3–5]. The physical content of that mechanism is a spin–spin four-fermion contact interaction: eliminating the non-propagating Cartan connection in favor of the fermionic spin current generates an effective interaction that grows with spin density and becomes repulsive well before the classical curvature singularity is reached, halting collapse. This is the same contact term this Note derives below, and the same sign.

Because the ECH bounce and Popławski's proposed link from bounce-scale torsion to a late-time vacuum energy [2, 3] are built from the same minimal field content, a natural question is whether the contact term responsible for the bounce can *also* source the dark-energy density observed today, $\rho_{\Lambda,\rm obs} \approx (2.25\,\rm meV)^4$. This Note answers that question systematically for the minimal-coupling ECH action: it merges the algebraic derivation and rigorous transparency theorem of an earlier companion paper [6] with the systematic no-go survey of a second companion [7] into a single, self-contained Classical and Quantum Gravity Note. Sec. II derives the contact term from the same Cartan equation that produces Popławski's bounce interaction and states its conditional-sign gap-equation result. Sec. III proves the perturbation-transparency theorem, the Note's sole Tier-I rigorous result. Sec. IV catalogs the fourteen mechanism-class constraints that jointly close the four enumerated dark-energy channels; Sec. V summarizes the two amplitude-budget route closures; Sec. VI states the

* houston@hubify.com

corrected operator-list argument. Sec. VII draws the positive/negative dichotomy this Note is organized around, and Sec. VIII concludes.

This is a channel-level assessment of the minimal-coupling theory, not an operator-level completeness theorem, and it makes no claim about non-minimal completions, propagating torsion, a dynamical Immirzi field, or quantum loops beyond the stated scope of each result.

II. THE CONTACT TERM: SAME MECHANISM AS THE BOUNCE

We use the first-order ECH action with constant Barbero–Immirzi parameter γ and a minimally coupled Dirac field,

$$S[e,\omega,\psi] = \frac{1}{4\kappa}\int\left(\epsilon_{IJKL} + \frac{2}{\gamma}\eta_{I[K}\eta_{L]J}\right) \times e^I \wedge e^J \wedge R^{KL}(\omega) + S_{\rm D}[e,\omega,\psi], \tag{1}$$

$\kappa = 8\pi G = 8\pi/M_{\rm Pl}^2$, mostly-plus signature $\eta_{IJ} = \mathrm{diag}(-,+,+,+)$, $\epsilon_{0123} = +1$. There is no independent torsion-squared term in Eq. (1); any four-fermion expression below is an on-shell result obtained only after the connection equation is solved. Varying ω_{IJ} gives the algebraic equation $\mathcal{Q}_\gamma(e^{[I} \wedge T^{J]}) = \mathcal{J}^{IJ}$ with $\mathcal{Q}_\gamma = \star + \gamma^{-1}\mathbb{1}$ invertible for every real finite nonzero γ [8]. For minimal fermion coupling the spin current is dual to $J_5^I = \bar\psi\gamma^I\gamma^5\psi$, and solving the connection equation and back-substituting (Freidel–Minic–Takeuchi [8], Eqs. 17 and 23, in the normalization bridge $4\pi G = \kappa/2$) gives the minimal axial–axial contact interaction

$$\mathcal{L}_{4\psi} = -\frac{3\kappa}{16}\frac{\gamma^2}{1+\gamma^2}\,(J_5^I J_{5I}). \tag{2}$$

This is Popławski's torsion-bounce interaction restricted to minimal coupling: it is generated by the identical elimination of the non-propagating connection, grows quadratically with the fermionic spin density, and carries the same repulsive sign that halts gravitational collapse in the black-hole-cosmology mechanism [1]. The Einstein–Cartan coefficient ($\gamma \to \infty$) is the maximal magnitude, since $\gamma^2/(1+\gamma^2) < 1$.

At finite γ the connection solution $e_I{}^\mu C_{\mu JK} \propto \left(\frac{1}{2}\epsilon_{IJKL}J_5^L - \gamma^{-1}\eta_{I[J}J^5_{K]}\right)$ carries two irreducible pieces: an axial part with coefficient $\alpha \propto \gamma^2/(\gamma^2+1)$ and a trace-vector part with coefficient $\beta \propto \gamma/(\gamma^2+1)$, ratio $\beta/\alpha = 1/(2\gamma)$; the tensor irrep vanishes identically for minimal coupling. This corrects an earlier catalog draft's "purely-axial" reading of the on-shell torsion, which holds only in the strict Einstein–Cartan limit $\gamma \to \infty$, not at the finite physical γ used elsewhere in this programme (theory-audit record [9]).

A. Finite-density benchmark and the NJL sign result

A deliberately elevated homogeneous normalization illustrates only the scale of Eq. (2):

$$\frac{\kappa n_\psi^2}{\rho_\Lambda} \simeq 3.6\times 10^{-69}\left(\frac{n_\psi}{100\,\mathrm{cm}^{-3}}\right)^2, \tag{3}$$

$\approx$68 orders of magnitude below $\rho_{\Lambda,\rm obs}$ even at an ISM-like number density. This coefficient-one benchmark omits the actual $3/16$ and finite-γ factors; number density also does not fix the renormalized composite $\langle J_5^I J_{5I}\rangle$, a vacuum stress tensor, or an equation of state.

In the declared direct-channel, hard-four-momentum-cutoff, standard mean-field NJL convention, Fierz rearrangement of Eq. (2) projects onto the scalar channel $G_s(\bar\psi\psi)^2$ with

$$G_s = -\frac{3\kappa}{16} < 0, \tag{4}$$

repulsive. The real homogeneous scalar gap equation of this convention therefore has no nonzero solution: this specific truncation admits no condensate. This conditional-sign result does not exclude other truncations, species structures, non-minimal couplings, or propagating torsion; the full regulated derivation is carried in the archived companion [6].

III. PERTURBATION TRANSPARENCY

The catalog's sole Tier-I rigorous result is the following. Consider the classical ECH action with an invertible tetrad, canonical scalar matter, and real finite nonzero constant γ, on a local patch with matched background, initial, and boundary data (standard falloff, vanishing first-order boundary variation). Solve the algebraic connection equation before expanding in perturbations. Then the reduced action on the zero-spin branch is the Einstein–scalar action, so the scalar equations and tensor evolution operators coincide with GR at every perturbative order for matched data. Equality of right- and left-helicity *solutions* additionally requires parity-symmetric initial data. The complex singular values $\gamma = \pm i$, nontrivial global/topological sectors, quantum loops or anomalies, fermion sources, non-minimal matter, a dynamical Immirzi field, and propagating torsion are outside this statement.

Proof. (1) A canonical scalar has zero spin density. (2) The zero source and invertible $\mathcal{Q}_\gamma$ give $e^{[I} \wedge T^{J]} = 0$; contracting twice with the dual frame shows the kernel is trivial for an invertible tetrad, so $T^I = 0$. (3) The connection reduces to Levi-Civita, $\Gamma^\lambda{}_{\mu\nu} = \mathring{\Gamma}^\lambda{}_{\mu\nu}$. (4) The Holst term $\frac{1}{2}\epsilon^{\mu\nu\rho\sigma}R_{\mu\nu\rho\sigma}(\mathring{\Gamma})$ vanishes identically on a torsion-free connection by the first (algebraic) Bianchi identity $R_{\mu[\nu\rho\sigma]} = 0$, which holds for any torsionless connection independent of metric compatibility. This

Bianchi-vanishing is a single-curvature identity, distinct from the two-curvature Pontryagin density $R\widetilde{R}$; no total-derivative argument is load-bearing. On an FRW background the tensor evolution operators are identical for both helicities, $\mathcal{E}_R = \mathcal{E}_L = \partial_\eta^2 + 2\mathcal{H}\partial_\eta + k^2$, so minimal ECH generates no GW birefringence, tensor chirality, or TB/EB CMB cross-power from parity-symmetric initial data, and the cubic action for ζ receives zero contribution from the Holst term in the stated domain. The theorem is an on-connection-shell equality of local classical reduced actions – not an off-shell equality of the original first-order ECH and Einstein–scalar actions – and its all-orders content follows from the algebraic Cartan constraint and the pointwise Bianchi identity, not an order-by-order calculation. A full self-contained statement and second-order Holst-term verification is carried in the archived companion [6].

TABLE I. The fourteen catalog entries – fourteen distinct mechanism-class constraints – on minimal-ECH dark-energy routes. B8 and B14 close the same observable channel (tensor parity violation) on disjoint matter branches and are logically independent (Sec. III).

#	Barrier	Src.	Blocks
1	Mass-Coupling Lock	A	Torsion as DE
2	Topological-Shift Duality	B	Pseudoscalar mass prot.
3	Scalar-Tensor Universality	C	Distinctive FRW geom.
4	Planck Suppression	D	Disformal coupling
5	Scale Separation	E	Global vacuum coupling
6	Attractor Sensitivity	F	IC transfer to DE
7	Parameter Immunity	G	Cyclic vacuum select.
8	Parity-Even Interaction	H	Tensor chirality (ferm.)
9	Liouville Conservation	J	Reversible selection
10	UV→IR Specificity	L	Generic/specific bridge
11	Decoupling Universality	L/M	Light gauge coupling
12	Vacuum Amplif. Ceiling	M	GW amplitude
13	Gravitational Democracy	N/O	Relics, baryogenesis
14	Perturbation Transparency	H	Tensor chirality (scal.)

IV. THE BARRIER CATALOG

Seven structural "foundations" (parameter-space, symmetry, and naturalness arguments; Barriers 1–7) and six observational branches (CMB, gravitational-wave, and relic-abundance channels; Barriers 8–14, with Branch H carrying two logically independent entries B8 and B14) together give fourteen distinct mechanism-class constraints that jointly close the four enumerated dark-energy channels (Table I). 'Distinct' and 'mechanism-class' mean no barrier is a logical consequence of another and each probes a separate physical failure mode; this does not assert disjoint assumptions, nor that each is an independently rigorous no-go theorem. Several barriers (B5, B6, B7, B10, B13) are general naturalness or classification arguments rather than sharp ECH-specific calculations, and B9 is an explicitly heuristic closure conditional on stated equilibrium assumptions that realistic quantum bounces can violate; B9 and B14 are never used as stand-alone closures of any route. The sole Tier-I leg among the fourteen is B14 (perturbation transparency, Sec. III), and it is Tier-I strictly on the zero-spin branch its hypotheses cover.

B1–B2 [R1, Founds. A–B]. Ultralight torsion modes require $g_{\rm eff} \sim H_0/M_{\rm Pl} \sim 10^{-61}$; reaching $g_{\rm eff} \sim 1$ needs $m_T \sim M_{\rm Pl}$, the standard CC hierarchy $\sim 10^{-122}$. In metric-affine gravity, mass protection $\Longleftrightarrow$ no geometric fingerprint: configurations protecting the pseudoscalar mass eliminate the geometric content and vice versa.

B3–B4 [R2, Founds. C–D]. Diffeomorphism invariance forces the torsion fluctuation to couple like any other FRW scalar, with no ECH-specific observable. Disformal torsion couplings are Planck-suppressed, $\mathcal{O}(10^{-122})$ at H_0.

B5–B7 [R3, Founds. E–G]; general naturalness arguments. No mechanism transfers the global vacuum integral across $N_{\rm tot} \approx 92$–94 e-folds; an inflationary attractor washes out bounce initial data while sensitivity to them destabilizes inflation; γ is fixed by the LQG area spectrum with no landscape for cyclic selection.

B8 [R1, Branch H]. $(J_5)^2$ is parity-*even* (product of two axial currents), so it cannot generate tensor chirality. Logically independent of B14: B14's hypotheses require zero spin density, precisely the condition B8's nonzero J^5 violates.

B9 [R2, Branch J]; heuristic. Phase-space conservation prevents irreversible post-bounce state selection under closed non-dissipative evolution with no particle production; scenarios that break these assumptions evade B9 and are constrained instead by B10–B11, so B9 is never used as a stand-alone closure.

B10–B11 [R3, Branch L/M]; B10 general naturalness. A Planck-to-H_0 bridge must be either generic or bounce-specific, never both; gauge fields decouple from the Planck-suppressed torsion sector universally, so ECH cannot preferentially couple to photons or dark-energy degrees of freedom without new non-minimal couplings.

B12 [R3, Branch M]. Bounce-epoch $\Omega_{\rm GW}^{\rm ECH} \lesssim (\rho_{\rm crit}/\rho_{\rm Pl})^2 \simeq 0.07$–$0.17$ using the LQG-bounce window; an order-of-magnitude ceiling, not directly comparable to the nHz PTA spectral density without propagating through the transfer function (deferred).

B13 [R4, Branch N/O]; structural observation. Torsion couples democratically to all spin-$\frac{1}{2}$ species and cannot preferentially source baryogenesis or relic abundances without species-dependent non-minimal couplings; logically independent of B11 (gauge universality) and B4 (coupling magnitude).

B14 [R2–R4, zero-spin branch, Branch H]. Sole Tier-I leg; Sec. III. Because its hypotheses assume zero spin density, it does not bear on R1; what it establishes for R2–R4 is that their classical zero-spin baseline is exactly inert, so any signal must come entirely from the quantum or non-minimal ingredient defining each route – bounded by Sec. V.

V. ROUTE 2–4: AMPLITUDE AND NATURALNESS CLOSURES

Route 1 (NJL contact) is closed by Sec. II: amplitude suppressed by $M_{\rm Pl}^{-2}$ and parity-even. Route 2 (one-loop graviton corrections to the Holst term) is anchored in the published one-loop renormalization of the Holst-plus-fermion sector [10] and closes against the observed birefringence amplitude with $\gtrsim$58 orders of suppression margin; its dark-energy leg is closed operator-level for constant Nieh–Yan coefficient (Sec. VI) and only at the naturalness level for a dynamical one. Route 3 (Immirzi running) is bounded by the integrated Benedetti–Speziale one-loop flow [11], leaving the contribution 61–67 orders below $\rho_{\Lambda,\rm obs}$. Route 4 (parity-odd CMB coupling via a spectator ALP or the fermion axial current) is not amplitude-suppressed: a free-normalization ALP fit reproduces both the WMAP+Planck birefringence signal $\beta_{\rm obs} = 0.342^\circ \pm 0.094^\circ$ [12, 13] (comparable to ACT DR6, $\beta = 0.215^\circ \pm 0.074^\circ$ [14]) *and* ρ_Λ, but minimal ECH derives neither the required ultralight mass $m_\theta \sim H_0$ nor the fitted coupling from first principles – it relocates rather than solves the cosmological-constant problem, so R4 closes as a naturalness/explanatory-deficit objection, not an amplitude exclusion.

The R1–R3 legs achieve mathematical amplitude closure (suppressed by explicit $M_{\rm Pl}$ powers and loop factors, many orders below observable thresholds regardless of normalization); R4 achieves naturalness/explanatory closure. These are logically distinct closure modes, classified separately above. The phenomenological parameter α/M that fits $\beta_{\rm obs}$ is therefore best understood as an R4-bounded coupling with the dark-energy density supplied by an unrelated sector, rather than by ECH-internal physics.

VI. THE OPERATOR LIST: RANK, NOT BASIS

The barrier catalog and route closures above bound the amplitude of representative parity-odd operators. A referee may reasonably ask whether the conclusion survives the full local operator space. Under the construction rule of zero additional derivative order, parity-odd ε contraction, full index contraction to a scalar density, and mass dimension exactly four, the admitted densities built from the tetrad, the torsionful curvature, the algebraically-fixed torsion, and the minimal axial current J^5 are spanned by a six-member list {O1–O6}. Independent symbolic adjudication [15] establishes that this list has **rank four** as a set of densities (two exact relations, O1 = O6 and O1 = $\frac{1}{2}$O4 − O2) and **rank two** modulo total derivatives – a deliberately redundant generating set of recognizable invariants, *not* a linearly independent basis; the list is asserted to span the rule-admitted space from the construction rule itself, not by exhaustive enumeration.

Every member reduces to one of three classes: an exact total derivative (O2, the Nieh–Yan density; O3, the Pontryagin density; and O1=O6 on the torsion-free branch), a Fierz-closed four-fermion contact term suppressed by $M_{\rm Pl}^{-2}$ (O5, the surviving vacuum-energy content, $\mathrm{O5}^{[4]} = -\frac{3}{2}\kappa(J^5 \cdot J^5)$), or identically vanishing. On the on-shell ECH branch the single-curvature pair (O1, O6) vanishes by the algebraic Bianchi identity on the torsion-free connection, exactly as in Sec. III; the ε-contracted torsion-square O4 was reported vanishing under a purely-axial reading of the on-shell torsion in an earlier catalog draft, but independent on-shell symbolic evaluation [9] shows the finite-γ ECH torsion carries a genuine trace-vector irrep alongside the axial one (Sec. II), so O4 is nonzero on the axial×trace-vector cross term and joins O5 in the Fierz-closed disposal class rather than vanishing outright. This does not change the closure: O4's surviving content is still $M_{\rm Pl}^{-2}$-suppressed and inside the same Fierz-closed basis as O5, so Route 2's dark-energy leg for constant Nieh–Yan coefficient is unaffected, only its stated mechanism is corrected.

VII. DISCUSSION

The two halves of this Note's title are the same calculation read in two directions. Eliminating the non-propagating ECH connection in favor of the fermionic spin current generates exactly one interaction – Eq. (2) – and that interaction is simultaneously (a) repulsive at high spin density, which is the physical content of Popławski's torsion-avoided singularity and bounce mechanism, and (b) parity-even, Planck-suppressed, and (on the zero-spin branch) classically transparent to every scalar and tensor perturbation – exactly the properties a viable dark-energy source cannot have. Minimal ECH is not a two-purpose mechanism with one purpose realized and the other merely unproven; the barrier catalog of Sec. IV shows that every distinct route from the bounce-scale torsion sector to a late-time Λ-like density is closed by a separate, named failure mode, and Sec. VI shows those failure modes are not an artifact of an incomplete operator enumeration.

This closure is specific to the ECH mechanism class under minimal coupling; it says nothing about non-minimal torsion couplings, propagating torsion, a dynamical Immirzi field, or the bounce's predicted galaxy-spin observable, which is tested independently by the DESI-scale chirality survey [16]. Nor does it bear on bounce cosmology generally: the matter-bounce $f_{\rm NL} = -35/16$ signature and NANOGrav-band gravitational-wave forecasts are properties of the matter-bounce class, not of ECH torsion, and are pursued in a separate research line [17]. What this Note closes is narrower and more useful to state cleanly: the specific claim that the same minimal

TABLE II. Evidentiary status of each leg: (I) rigorous within stated scope, (II) structural argument, (III) ansatz-level dimensional estimate. No leg is claimed more strongly than recorded here.

Leg	Status
Perturbation transparency	**(I)** Rigorous within stated scope.
R1 (NJL contact)	**(II)+(III)**: parity-even structure is algebraic; $M_{\rm Pl}^{-2}$ suppression is an ansatz-level estimate.
R2 (one-loop)	**(III)** birefringence leg; **(II)** dark-energy leg (both branches).
R3 (Immirzi running)	**(II)+(III)**: mass-dimension lock is structural; the chiral-count bound is a loose Tier-III ceiling.
R4 (CMB/spectator-ALP)	**(II)** structural naturalness objection, not an amplitude exclusion.

spin-torsion sector producing the ECH bounce also supplies the observed dark-energy density.

VIII. CONCLUSIONS

Minimal Einstein–Cartan–Holst torsion, under strictly minimal fermion coupling, generates one physical interaction: a spin–spin four-fermion contact term, repulsive at high density, algebraically identical to the mechanism underlying Popławski's torsion-avoided singularity and black-hole-cosmology bounce. This Note shows, at channel-amplitude granularity and within a stated minimal-coupling scope, that the same interaction cannot additionally source the observed late-time dark-energy density: its NJL scalar projection admits no condensate in the declared truncation (Sec. II), it decouples identically from classical perturbations on the zero-spin branch by a rigorous Bianchi-identity theorem (Sec. III), fourteen mechanism-class constraints jointly close all four enumerated dark-energy channels (Sec. IV–V), and a rank-four spanning list of the admissible dimension-four parity-odd operators shows this conclusion is not an artifact of an incomplete operator basis (Sec. VI). The positive result – what minimal ECH does for the bounce – and the negative result – what it cannot do for dark energy – are two readings of the same algebraic elimination, and this Note's contribution is to state both precisely, with the closure scope and evidentiary strength of every claim recorded explicitly rather than implied.

Data and Code Availability

The symbolic verification scripts supporting the transparency theorem, the on-shell torsion-irrep computation, and the operator-list adjudication are publicly available in the companion papers' repository, `https://github.com/Hubify-Projects/bigbounce`:

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The earlier version of the algebraic torsion-elimination derivation and its regulated NJL check is archived, with its own verification scripts, as an immutable deposit at doi:10.5281/zenodo.21481838 (CC-BY-4.0) [6]. This Note supersedes that paper and the companion no-go survey [7] as the single submission target for the closed ECH dark-energy line; both source manuscripts remain archived and are not independently resubmitted.

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